

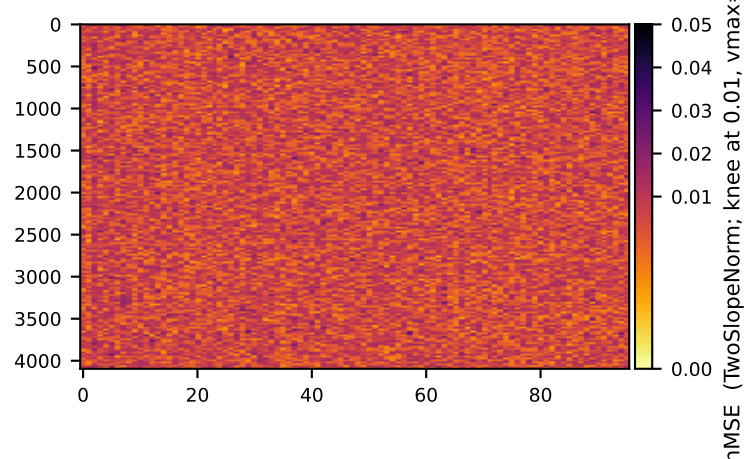
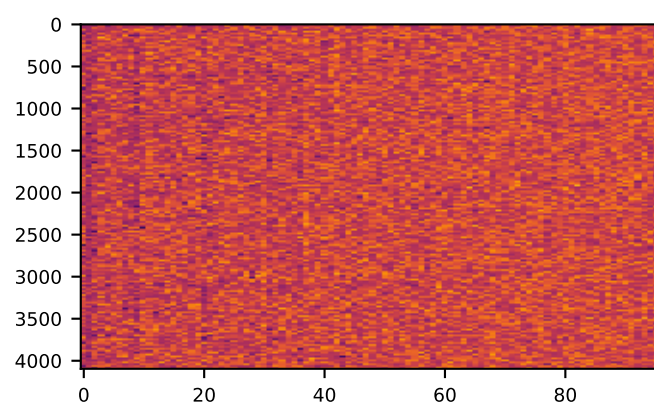
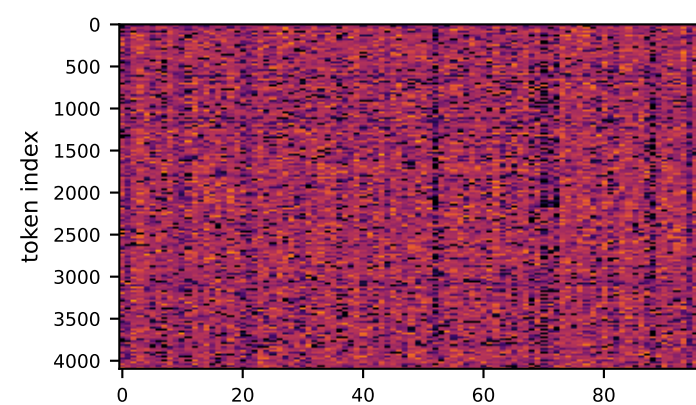
Activation quant diffculty after rotation — gr00t + pi0.5 combined
Channel axis binned by block_size=64 (= int4 quant block). Each cell = per-(token, block) normalized MSE
 $\|X_{t,b} - Q_4(X_{t,b})\|^2 / \|X_{t,b}\|^2$ under MSE-optimal int4 scale. Bright yellow = small nMSE (easy); black = hard.

raw (no rotation)

DuQuant SVD

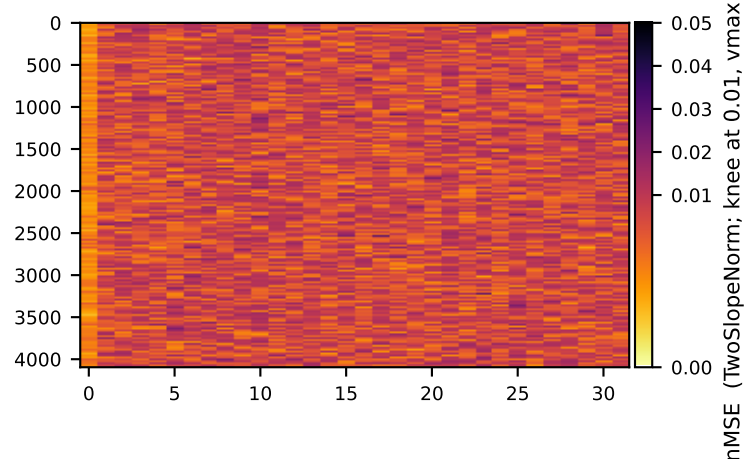
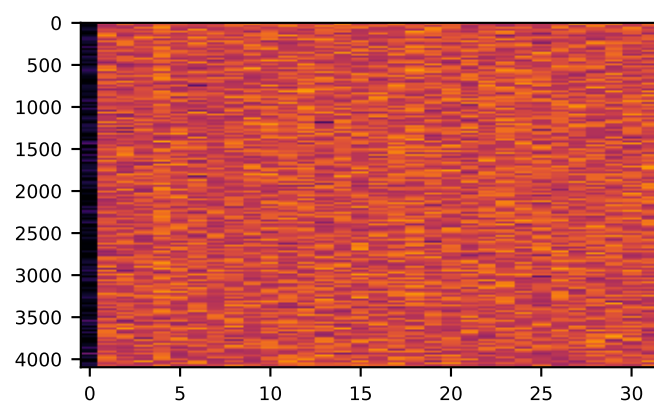
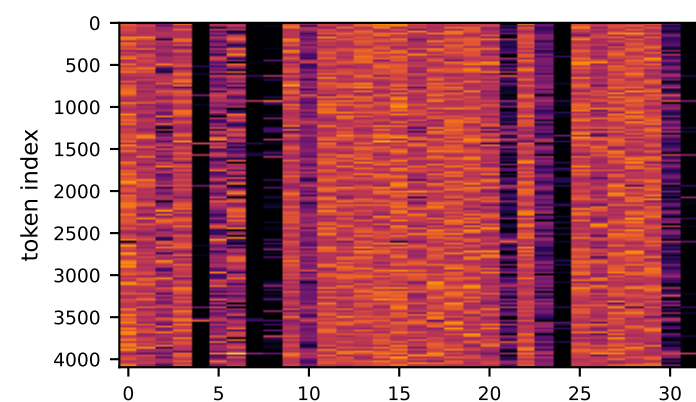
SVD-Hadamard (A2-lite)

gr00t LLM.L02.down_proj — pathology (raw channel-skew 48000×)



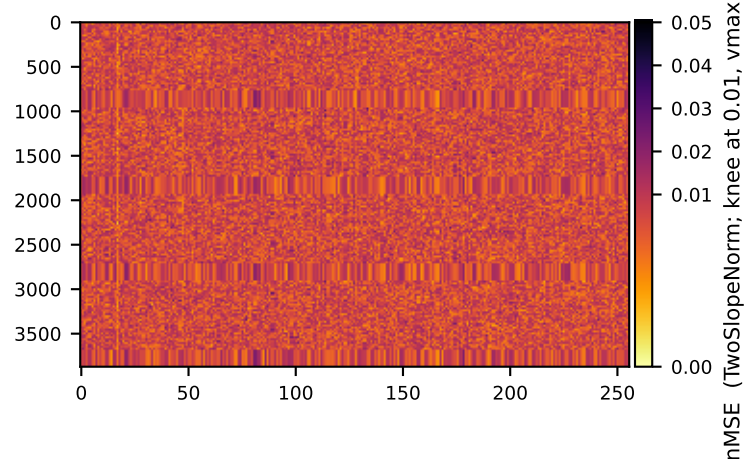
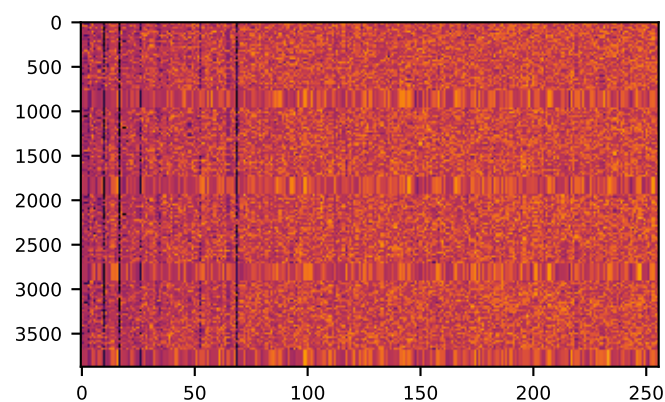
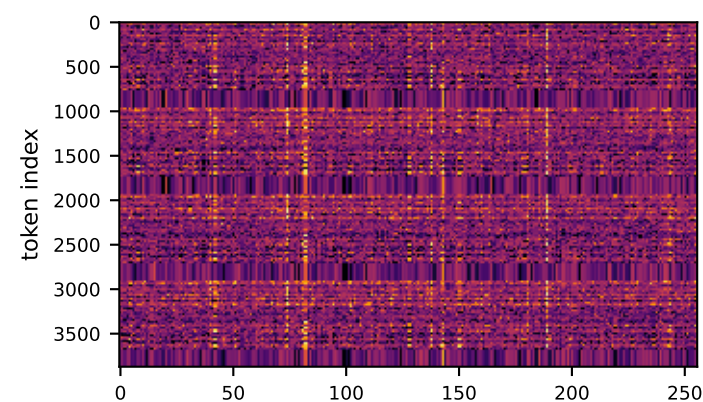
nMSE (TwoSlopeNorm; knee at 0.01, vmax=0.05)

gr00t LLM.L02.gate_proj — typical MLP (raw skew 59×)



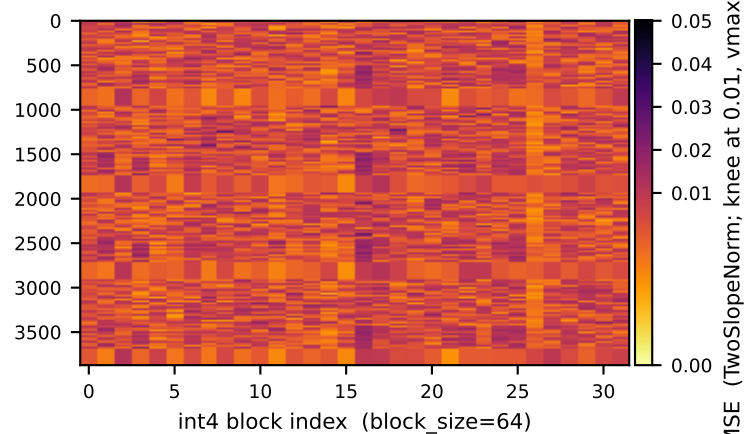
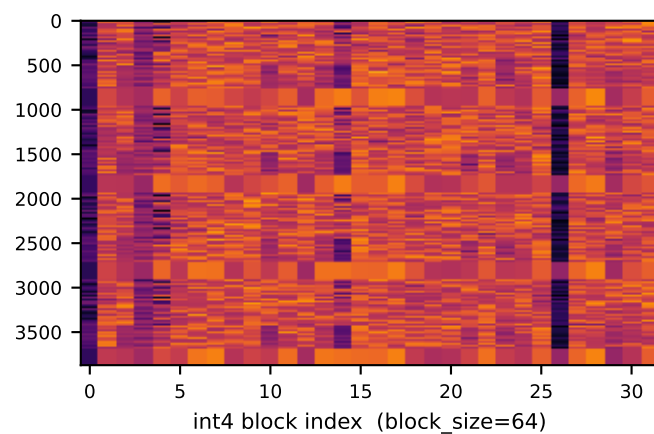
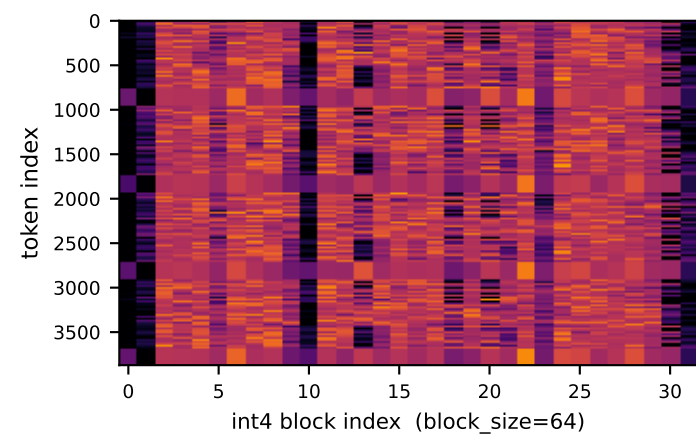
nMSE (TwoSlopeNorm; knee at 0.01, vmax=0.05)

pi0.5 paligemma.L14.down_proj — pathology analog (raw skew 156×)



nMSE (TwoSlopeNorm; knee at 0.01, vmax=0.05)

pi0.5 paligemma.L02.gate_proj — typical MLP (raw skew 12.4×)



nMSE (TwoSlopeNorm; knee at 0.01, vmax=0.05)